Ragas and Microtonal Articulations in Indian Classical Music

How microtonal articulation shapes Raga Identity and Emotional Perception

Research Report

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A Note on Terminology

Sa Re Ga Ma Pa Dha Ni - the seven notes of Indian classical music, relative to the performer's chosen tonic. Roughly equivalent to Do Re Mi Fa Sol La Ti in Western solfege.

Shuddha - natural or unaltered form of a note.

Tivra - sharpened by a semitone; only Ma has a tivra form.

Komal - flattened by a semitone; Ati-komal- further flattened beyond standard komal (a microtonal shift in pitch position itself).

Raga - a melodic framework defined by its note set, characteristic phrases, ornamental vocabulary, and emotional character. Two ragas can share the same notes and still be entirely distinct.

Gamaka - heavy oscillatory pitch movement; a key carrier of melodic identity.

Meend - a continuous glide between two notes following raga-specific rules.

Andolan - a slow, raga-specific oscillation on a sustained note, distinct from vibrato in that its depth and range are defined by the raga.

Pakad - the signature phrase of a raga, encoding its most characteristic gestures in concentrated form.

Research Question and Context

The central question driving this research is: how is the essence of a raga expressed in instrumental Indian classical music? Beneath that sit several related inquiries. How do ornamentation and vocal-style articulation contribute to expressing raga identity in instrumental performance? What specific ornamentation techniques define a raga, and how does their presence or absence change perception? Why do ragas built on nearly identical note structures feel fundamentally different in practice? And to what extent is the perception of raga identity culturally conditioned rather than universal?

These questions arise from a genuine gap between how raga is theorised and how it is experienced. Depending on the source, raga identity gets described through rasa, through melodic grammar and rule-sets, or through the quality of movement a performer brings to the notes. These are not minor differences in emphasis. They reflect fundamentally different intuitions about what a raga is. Performers and trained listeners can sense almost immediately when a raga is being expressed convincingly or when something has gone wrong. Explaining exactly why, in terms that hold across different contexts and listeners, is considerably harder (Deshpande, 2024).

This research situates itself at that gap. Rather than describing raga as it functions within tradition, it asks what happens at the edges: how far can the defining characteristics of a raga be transformed before it becomes unrecognisable, and which elements are genuinely that hold musical value?

Methodology

This research employed three complementary strands of inquiry. The first was a structured literature review examining the primary theoretical frameworks for raga identity, drawing on Kelkar and Indurkhya (2014), who propose that raga is better understood as continuous motion through pitch space than as a sequence of discrete notes; Ganguli et al. (2010), who demonstrate computationally that ornaments carry structurally integral pitch curve signatures; Nuttall et al. (2025), whose ornament recognition work models continuous pitch contours as primary data; Sarkar et al. (2021), whose statistical modelling begins to formalise the topology of melodic motion; and Deshpande (2024), who identifies how shifting conditions of transmission and reception affect raga perception.

The second strand was two structured qualitative interviews with practising musicians, each approximately ninety minutes and including live demonstrations. Mr. Paulson K J, Hindustani Classical Sitarist from Thrissur, Kerala, with 30 years of experience, was interviewed on raga identity, the constitutive role of meend and andolan, and perception across listener backgrounds. Mr. Abhay Nayampally, Carnatic Classical Guitarist, also with 30 years of experience, from Bengaluru, Karnataka and an ardent senior disciple of the late Mandolin Vidwan U. Srinivas, was interviewed on the Carnatic gamaka system and the experience of producing continuous pitch articulation on a fixed-fret instrument.

The third strand was an audio-visual installation, Antara - Between the Notes, presented at Carnegie Library, Dun Laoghaire, 2026. The installation placed two versions of the same melodic material in opposing halves of a single space: one on bansuri flute with full microtonal ornamentation, the other on piano in equal temperament with none. Three ragas were used: Darbari Kanada, Raga Jog, and Raga Kafi. Twenty audience members completed structured response forms, providing perceptual data on how the two versions were experienced.

Findings from the Interviews

Both interviews consistently supported the argument that microtonal articulation is constitutive of raga identity rather than ornamental, while surfacing positions that add nuance to that claim.

Paulson described raga identity as residing primarily in the pathways of movement between notes. His demonstration of Darbari Kanada was particularly illuminating. Playing the characteristic phrase on komal Ga first with the slow andolan that defines the raga, then with the same note held clean and still, produced a contrast immediately audible even to untrained listeners. The andolan on Darbari's komal Ga is not a standard flattened third. It occupies a specific ati-komal position and oscillates so deeply it nearly reaches Re. Removing it collapsed the raga entirely. This aligns directly with Kelkar and Indurkhya's (2014) motion-based framework and with Ganguli et al. (2010), who demonstrate that ornamental gestures carry structurally integral pitch curve signatures distinct to each raga.

Paulson also argued that raga identity is medium-independent, emerging through the interaction of pitch, motion, and time rather than through any particular instrument. This has direct implications for the installation methodology: if raga identity is medium-independent, the perceptual difference between the bansuri and piano versions cannot be attributed to timbre alone. It must be attributed to what the piano cannot do, which is produce continuous pitch movement.

Abhay's interview introduced a complication the Hindustani-focused literature does not address. As a Carnatic musician on a fixed-fret guitar, every gamaka requires physically bending strings away from their equal-tempered positions. His observation that this resistance sometimes becomes part of the expression raises a question Ganguli et al. (2010) cannot capture in a pitch curve: whether the effort of producing a gesture contributes to its emotional quality.

The Hamsadhwani cross-traditional demonstration, performed by both musicians, revealed that the same pitch set produces entirely different ornamental vocabularies in Hindustani and Carnatic contexts, yet both remain recognisable as Hamsadhwani. This suggests raga identity has a stable core that persists across different ornamental languages, while the specific gestures through which it is expressed are tradition-specific. This layered understanding is consistent with Deshpande's (2024) observation that conditions of reception shape what listeners can perceive without determining whether the raga is present.

Findings from the Installation

Twenty responses were collected across three ragas: Darbari Kanada (three responses), Raga Jog (eight responses), and Raga Kafi (nine responses). Seventeen respondents identified as musicians and three as non-musicians. Age range was 25 to 66 years, with an average age of 43, with eleven female and nine male respondents.

The most significant finding was the near-unanimous identification of the flute side as more emotionally expressive: eighteen out of twenty respondents chose the flute side, one chose keys, and one felt both equally expressive. Since the only variable separating the two sides was the instrument and its capacity for continuous pitch, this result points directly to the role of microtonal articulation in generating emotional response, supporting Kelkar and Indurkhya's (2014) position that continuous pitch movement carries the identity of the music.

The pitch bending data reinforced this. Nineteen out of twenty clearly noticed pitch bending on the flute side. On the keys side, nine noticed none at all, seven noticed a little, and three said yes clearly, likely registering equal-tempered sustain as movement, itself a perceptually interesting result.

The structural perception data produced the most analytically significant finding. Thirteen respondents described the flute as Alive and Breathing. The keys side was described as Mechanical and Fixed by seven respondents, and as Strange and New by six. The Strange and New response is the most revealing. If removing microtonal ornamentation produced something Western-sounding and familiar, trained musicians would be expected to find the keys side more comfortable. Instead they found it foreign. This suggests equal temperament applied to Indian classical melodic material does not produce a Western equivalent. It produces something belonging to neither tradition, consistent with Nuttall et al.'s (2025) finding that Indian classical ornamental pitch contours are not reducible to Western analytical categories.

The one-word responses made divergent perception most visible. Flute words included: Emotional, Moving, Spiritual, Transcendent, Human, Breath, Fluid, Whispering, Ethereal, Mysterious. Keys words included: Mechanical, Anxious, Static, Harsh, Distant, Sharp, Clean, Experimental, Transient. The two vocabularies did not cross at all. The three non-musician respondents did not differ significantly from musicians, all identifying the flute side as more expressive and clearly noticing pitch bending, suggesting the perceptual difference is accessible across different levels of cultural familiarity.

Conclusion and Future Directions

The evidence gathered across three methodological strands converges on a consistent position: microtonal ornamentation in Indian classical music is not decorative. It is constitutive. The identity of a raga, its emotional quality, its perceptual character, and its capacity to be recognised, rests significantly on the specific gestures through which its notes are approached, held, and connected. Removing those gestures does not simplify the music. It produces something that trained and untrained listeners alike find foreign rather than familiar.

This is supported by the interview data, where performing the same phrase with and without characteristic ornamentation produced an immediate and audible collapse of raga identity; by the installation data, which showed near-unanimous divergence in emotional response between the two versions; and by the literature, which has moved progressively toward a motion-based understanding in which pitch trajectories rather than pitch positions carry identity (Kelkar & Indurkhya, 2014; Ganguli et al., 2010).

What this research cannot yet resolve is the precise boundary between tradition-specific ornamental vocabulary and underlying raga identity. The cross-traditional demonstration suggested that raga identity has a stable core persisting across different ornamental languages, but the research did not isolate what that core consists of. Future work could remove individual ornamental elements one at a time to identify which gestures are load-bearing and which can be altered without collapsing recognition.

The gap that remains most pressing is between the scientific and aesthetic dimensions of this question. The audience response data and interview material describe a perceptual and emotional reality that the computational literature has not fully captured. Brain mapping studies and more controlled psychoacoustic experiments, in which variables such as timbre, rhythm, and familiarity are more precisely isolated, would provide substantially more conclusive findings on the physiological basis of microtonal perception. Until that bridge is built, the scientific and aesthetic accounts of raga identity will continue to describe the same phenomenon in languages that do not fully translate into each other.

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Appendices

Appendix A — Interview Recordings Combined video documentation, shortened into the topics relevant for the research paper, of the two researcher interviews conducted for this project: Mr. Paulson K J, Hindustani Classical Sitarist, and Mr. Abhay Nayampally, Carnatic Classical Guitarist. Interviews include live musical demonstrations referenced in Section 3.

https://youtu.be/3jtZpy1BjrU

Appendix B — Audience Response Form The questionnaire distributed to installation visitors at Carnegie Library, Dun Laoghaire, 2026.

https://drive.google.com/file/d/1a0RUwmbGA2w_p9q14AHp-pIF4YvTM0OI/view?usp=drive_link

Appendix C — Audience Response Data Raw data collected from twenty respondents across three ragas: Darbari Kanada, Raga Jog, and Raga Kafi.

https://docs.google.com/spreadsheets/d/1mE983PN0-g5YNsk277pmZ4_Hvrnwvt2a/edit?usp=sharing&ouid=111304077546195261519&rtpof=true&sd=true

Appendix D — Installation Documentation Photographs and technical notes from the Antara installation, Carnegie Library, Dun Laoghaire, 2026.

https://drive.google.com/file/d/17NpteMVwTNsEMFkHlPyaOSuOwK_v-t06/view?usp=drive_link

Appendix E — Website A combined repository with artefacts and information of research

https://abhinavkjayan005.github.io/Antara/